

Heavens light is our guide

Rajshahi University of Engineering and Technology

Electrical & Computer Engineering



ECE 3100:Software Development Project-2

Software Project Report

Daily Weather app

Submitted To

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Daily Weather App

Project Type: Software

Framework: Flutter

Programming Language: Dart

Backend and Frontend Communication: REST API

We made an app giving importance to functionality more than User Interface. We were able to bring these features in our weather app.

Features

1. Daily and Hourly Forecast

The data of daily and hourly forecast will be available in the app.

2. Search Bar

A search bar will be available from where we can know the weather information of various cities.

3. Celsius and Fahrenheit Conversion

If a user doesn't want to see weather information in Celsius unit, then he or she can go to the settings option to alter it to Fahrenheit scale.

4. Current Location Access

When the app is installed for the first time, the app will require the data of the user's current location using the Geolocator plugin, for which a permission option will pop up. Then the user can access the home screen where information like temperature, division, city and other details will be seen.

5. Five Days Weather Forecast

A horizontal ListView will be available which will provide information for every 3 hours of the next 5 days.

6. City Suggestions

Initially, there will be some suggestions of cities.

7. Animated Background

There will be usage of static images in the background which will be in motion to make the app more realistic.

Why did we make Weather App?

The question is why we made a weather app whereas in every smartphone a weather app is installed by default or weather information can be seen directly from websites. Why would anyone bother to download another weather app?

Mainly, our main focus was on REST API. Nowadays, the software industry uses REST API a lot for their applications as they need to be connected with servers.

With REST API, we can send data to the server, retrieve data from the server, update existing data, delete data, and perform authentication operations.

Similar functionalities can be achieved using Firebase too. However, not everyone uses Firebase. Many companies create their own backend systems and provide services through REST APIs.

Many companies can buy a domain and then build their backend using Laravel. For advanced-level development, backend systems can be created using Node.js or Spring Boot.

However, mobile app developers prefer using REST APIs because there is no extra hassle of creating backend infrastructure when a service provider has already created the backend system for us.

We will just grab that data from there and display it in our app. There is no need to be concerned with handling the weather database directly.

REST API will be provided to us and we will just call that API to bring the required data.

HTTP Library Usage

We used the HTTP library for making API requests. There are more popular libraries available such as Retrofit, Dio, and other networking-related libraries.

As the HTTP library is Dart's official package, we used it for this project.

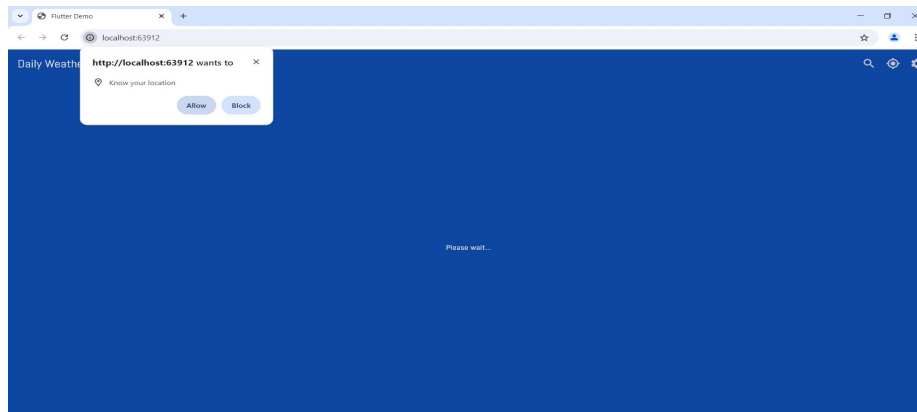
Now we will describe the features of our application in detail.

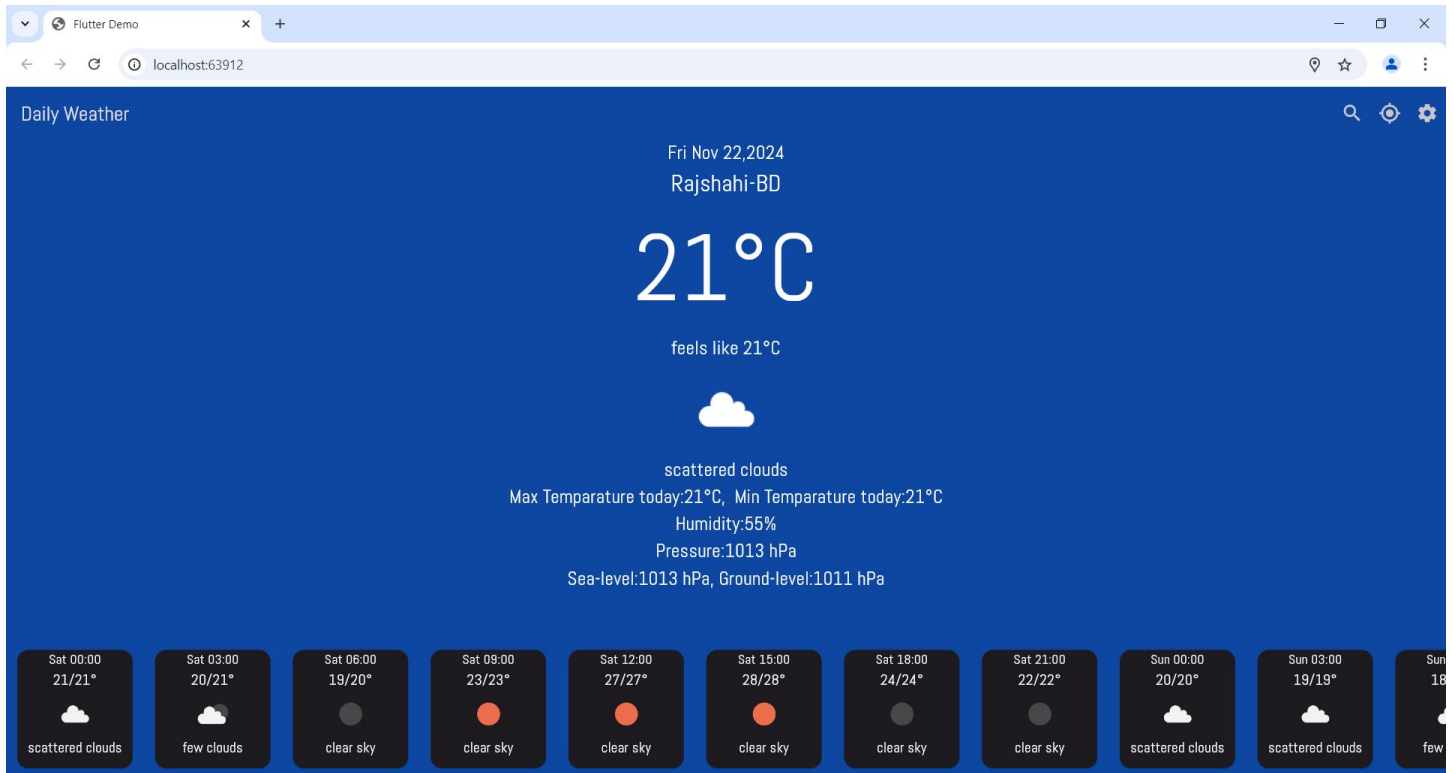
Description:

How did we show daily data and next 5 days data from json data to UI element?

As weather changes constantly, the data also changes for which we will use rest api. It's like a network or online server from where we send get request to that online database, we get

response as a json.json is key value pair.so in the json all over information about forecast and daily weather will be there.this much we did in browser.but we want to call it from our project.so that's why we need to convert those json to dart files.for forecast we turned the json into ForecastWeather class and for current weather, we turned the json into CurrentWeather class.The corresponding class of objects in the json are built in these two classes.now we will throw get request from our project for which we created two methods like getCurrentdata() and getForecastdata() in weather_provider.dart file.as for now, we did work on a certain latitude and longitude so we will call these methods through providers under didchangedependency() method in weather_home file.to show these informations in the ui, we will make two sections.current weather section and forecast weather section and then will call from weather-home.dart file.in currentweather, there will be information like country,city,temperature, feels like, date etc.as weather-home is our main page from where everthing is being called.so under inside body section of scaffold, we will call these section inside column widget.before calling these methods inside the column method, this two section need to be fed the information from the getter method of weather_provider.so that's why we wrapped the column method inside the consumer widget which will consume weather_provider and throw an object of this to column.column's sections will consume these informations.





Settings page and conversion of unit:

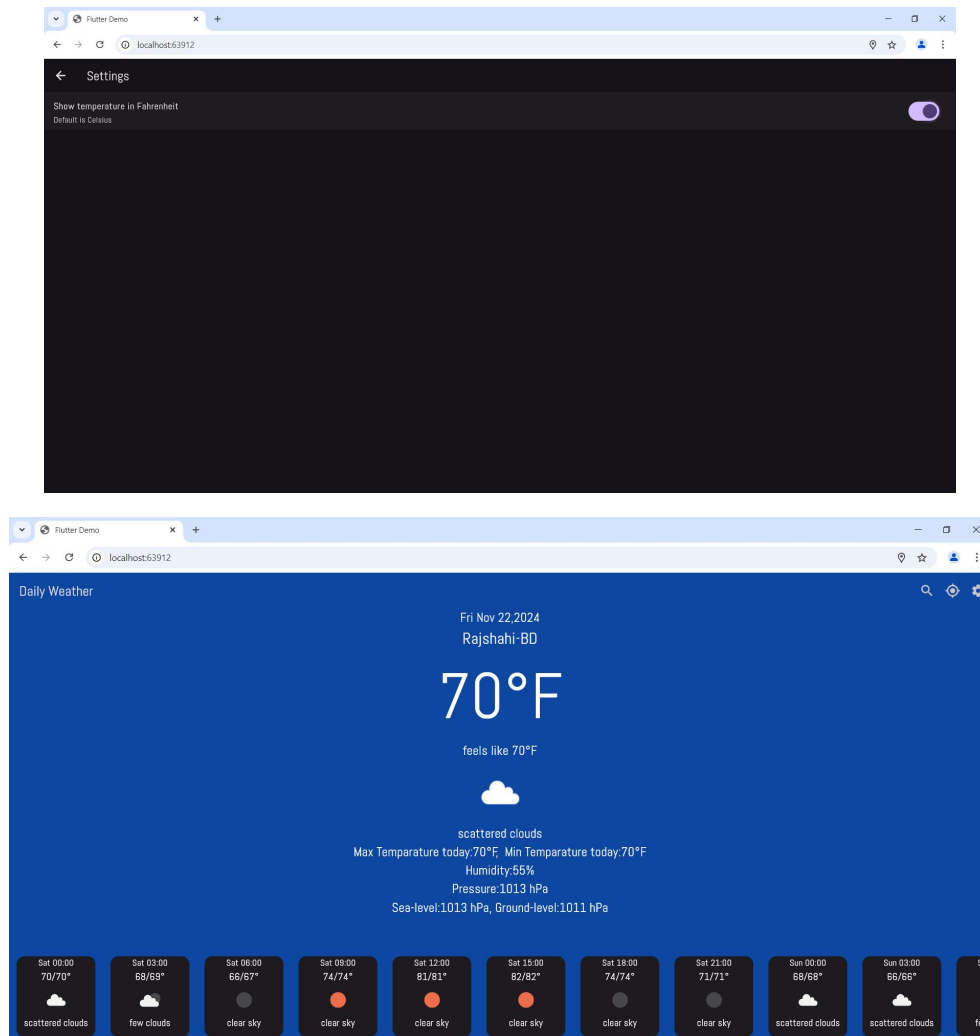
In weather provider, the unit is by default set to metric unit. In order to change that, we need to work on settings page. There will be a button which will be brought by `switchListTile` widget of dart. This widget has a `switch` property which will turn on and off on the basis of Boolean value. Without Boolean value, the switch will move right and left but there lies no functionality. When user turn on the switch, `onChange` method will be called where the Boolean values are stored. Then it will be stored in `ison` variable of this `switchListTile` widget. This is how it works.

Switch on=weatherprovider unit will be change to imperial.

Switch off=weatherprovider unit change back to metric.

We are using weatherprovider object everywhere because it sustains all the features that needs to be introduced in the app. Now how the onchange method stores Boolean value? when we are turning the witch on or off, `setTempUnit` method from provider is called to settings page. `setTempUnit` has switch value as argument. So, if the value is true, these Boolean value will be sent to `onchange` method of `switchListTile` eventually setting the value in `ison` and the functionality is built in this way. Another thing is, our work is not done after calling the `setTempUnit` method of provider to settings page's `switchListTile` method. Right after calling it, we should call `getData` method provider. The open weather server will get the request that right now I need the data in Fahrenheit unit. the data will be sent to `getData` method. Thus our

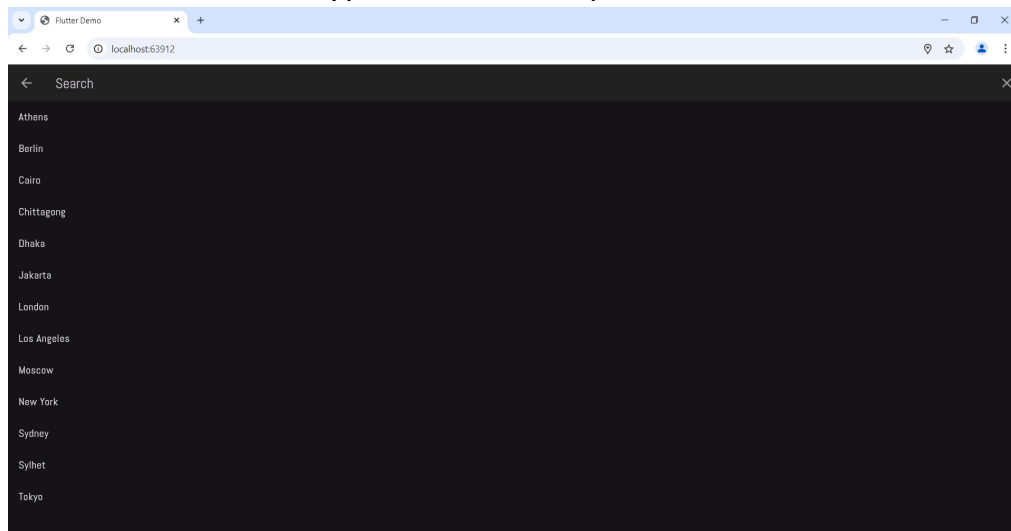
UI element will get the updated value or Fahrenheit value. The json value must be stored in shared preference. If we don't do it, everytime I leave the setting page even if I turned on it will go to off state again showing us previous value in UI element.



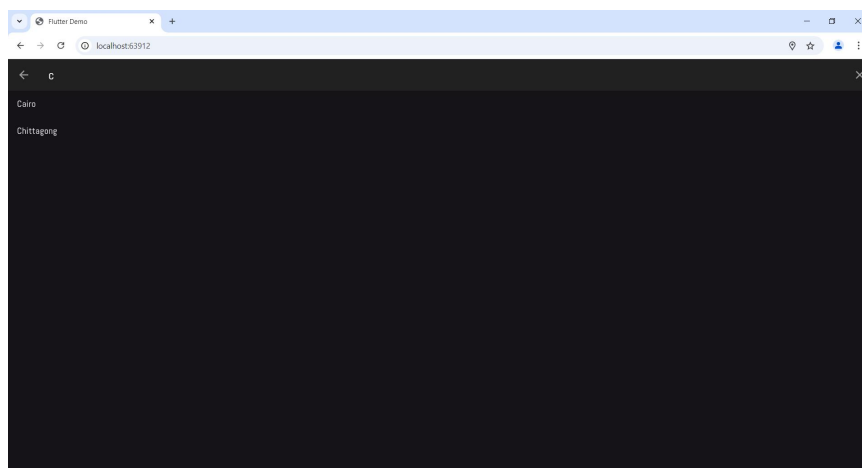
Search Delegation (search bar, suggestion, query data get):

Through search interface, user will be able to see the weather of other cities. For that, we will use flutter sdk's default search interface (showSearch) method in the onPressed method of search icon in weatherHome page. There is a searchDelegate class in the showSearch method. As it is an abstract class, we have to make a child class of it then send to the showSearch method. From taking user query to save it, returning query result, everything is handled by searchDelegate class. The child class of searchDelegate is _citySearchDelegate, a private method which will be called outside of main body. As it is a child, it must extend parent class, searchDelegate. So as a result four methods will be taken from parent class which will throw individual UI components.

- **buildAction:** it is used for search box or to push search page. IN search box, People type which is set in the query later on. If clear button is tapped, whole query will be erased.
- **buildLeading:** the back arrow used here indicates how to go to previous page. It will use close method of parent class as goes back to previous state.
- **buildResult:** after searching, through Text widget we will show the result. Apart from city suggestion list, any other unknown but valid city outside of the list will be shown through this method.
- **buildSuggestion:** during user is typing something, a list of suggestion will be given. He can choose from there. This list comes from constant.dart file where cities are stored in static list. After choosing city, search page will close. The result will be returned to this buildSuggestion method. One keypoint is the complete list will not be shown.



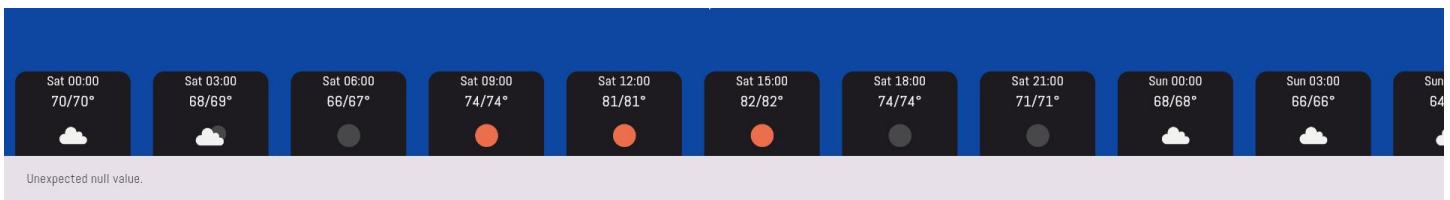
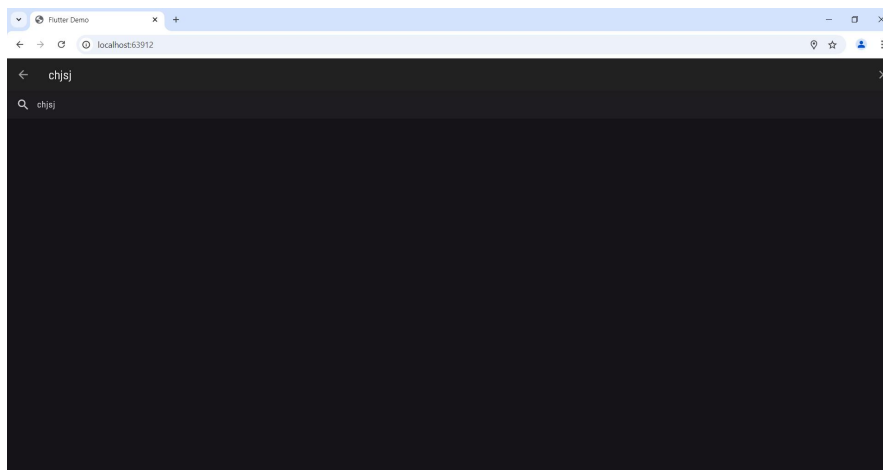
Upon the first letter that user enters, query list will be shown. For that purpose, we have returned filtered list. After choosing particular city from the suggestion, close method will come from parent class, searchDelegate which will shut down the search interface. The result for the query stored in the index of filtered list will be shown to UI element.



The city we got from the search interface, all information of it must be shown in weatherHome. Even if we close the search interface, the keyboard interface doesn't close which may create keyboard renderflex issue. So, we need to close the keyboard too.

Convert city to coordinates:

We used latitude, longitude to grab weather data. But this time we will use geocoding library which will turn the street address to lat or long or vice versa. The city which's lat., long., we want will be set into weatherProvider class's lat., long. The method we will use for such conversion of address is `convertCityToCoord` where city will be passed as argument. We have to do all over procedure in try and catch block. Because if user types a city which doesn't exist at all,



then an exception will be handled by catch block. That means error will be handled by catch block. We have to print the error to see it in the console in details otherwise it's not possible to see full error description in UI element. In try block, we will use geocoding api's `locationFromAddress` method. There can be times when our city is valid but during parsing the city, an empty list is returned. We will work when the list is not empty. Else there will be message like could not find location or error. The list can return with a couple of famous location of a city. We showed only first city of the location list which will be set as `newLocation`. Along with this location's lat., long. will be set in the method named `setNewLocation` inside of the `convertCityToCoord` method. Right after that, `getData` method will be called which will send request to server to provide the corresponding data against the location's latitude and longitude. Now this method will be called under `citySearchDelegate`

method of weatherHome class for condition like the value which is query ultimately should not be null.

Back to previous location:

If we want to go back to our location we have to close the app again. Instead we introduced a separate my location button which will give data according to user's actual location. In this case, we called getLocation method in My location button.

Parallax effect:



We used four static images which will slowly move from right to left. We won't use those kind of animation which will lower down our app's speed. First we took a sized box height of 200 then a stack widget because animation position has to be given. This stack's child is SizedBox which is of 2000 width. This Sizedbox has child again named opacity. Opacity has child named row on which animation position is used. If left value will go in negative and the sizebox will move to right. So to keep this movement consistently, we have to change left value again and again. For this, we introduced a method named startMovement under didChangeDependency method. This startMovement will change the offset value.

$$\text{Left} = \text{offset} + \text{screenwidth}$$

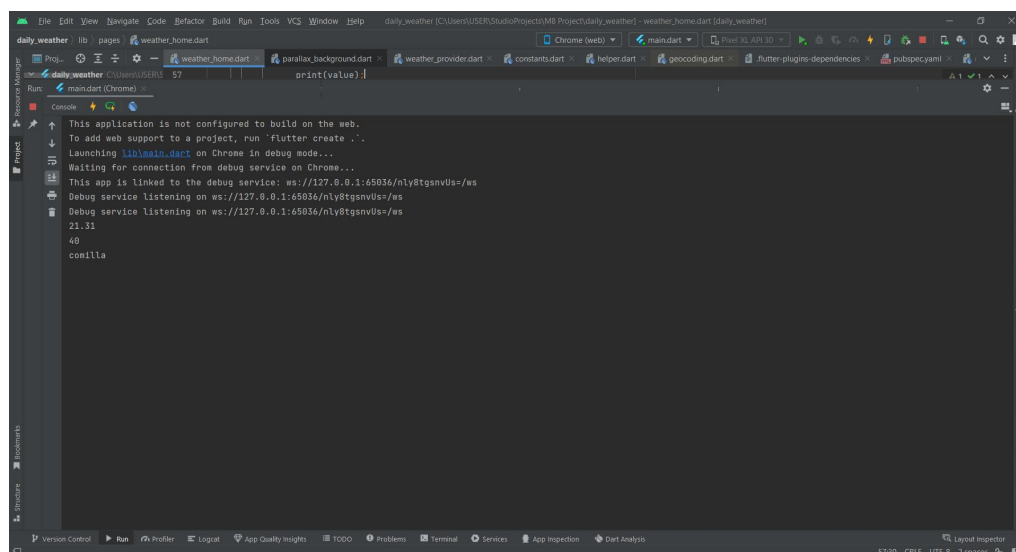
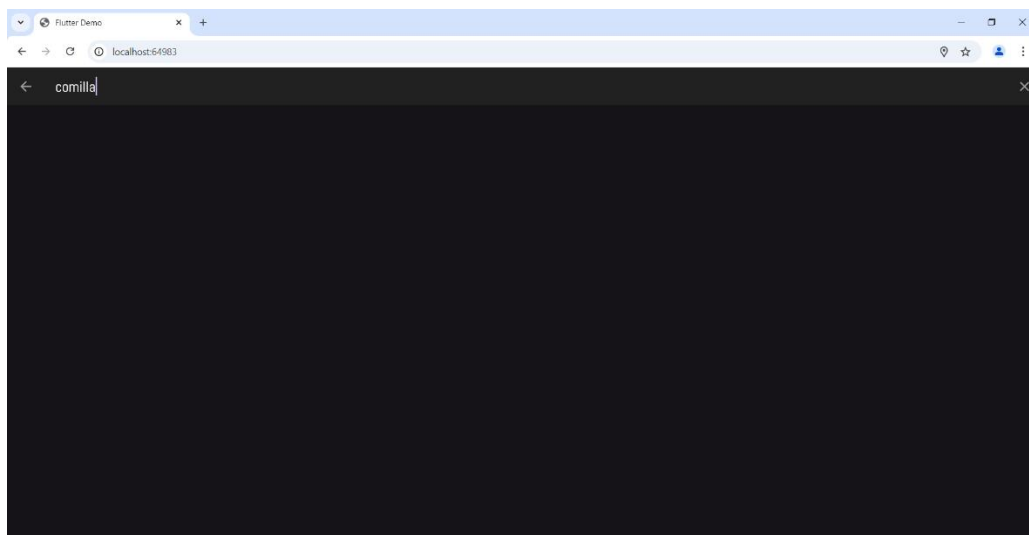
The moment sizebox will be at the right most position that means it is about to go out of the screen, we have to lessen the opacity of sizebox from 0.5 to 0 because there will be blink if the

sizebox moves to the rightmost position again to keep the movement from right to left. Once it is set to starting or right position, the opacity is changed back to previous value, 0.5. After every 50ms, startMovement method is called. That means, sizebox is shifted to left by 0.5. In this way, offset value for left will be -0.5, -1, -1.5, -2.... This is the theory why left is getting minus value. We could have done the same thing with animated api.

In weatherprovider class, we called this parallax method before the Text widget under stack as we want to show text widget first beneath parallax background will be shown.

Problems that was faced in the project:

- As openweather's free api key was used, it doesn't give access all the information of all cities.
- Even though our city is valid, city could not be parsed due to limitation of api usage for which the address list that is needed in convert city to coordinates returns empty.



Here even though our city is valid, the city name only returned as string.it could not parse relevant information to comilla.That's why every time, we entered a valid city, it gave us unexpected null value.This issue maybe solved with the paid version of openweather api site.

Conclusion:

Api calling, receiving response, json to model class conversion. These all are common thing to do when we want to work with api for our app. Specially when we work with json. There will be times that the response may come in XML mode.This is also easy to parse. Here, key value pair is followed too like json. So , the main learning was to learn how to use api. With this, we can build apps like earthquake app using earthquake api USGS, movie app, news app.